

Indices

Chapter 1 Indices 1

$$2 \times 2 \times 2 = 2^3$$

← Index
← Base

$$\begin{aligned} (-2)^2 &= -2 \times -2 = 4 \\ (-2)^3 &= -2 \times -2 \times -2 = -8 \\ (-2)^4 &= -2 \times -2 \times -2 \times -2 = 16 \end{aligned}$$

Index Law 1

$$a^m \times a^n = a^{m+n}$$

Index Law 2

$$a^m \div a^n = a^{m-n}$$

Index Law 3

$$(a^m)^n = a^{m \times n}$$

Zero Index

$$a^0 = 1$$

Negative Index

$$a^{-m} = \frac{1}{a^m}$$

- 1 Write $a \times a \times a \times a \times a \times a \times b \times b \times b$ in index form.

$$a^6 \times b^3$$

- 2 Simplify $7a^0$.

$$7 \times 1 = 7$$

- 3 Simplify $\frac{16x^2y^2}{4x^2y^4}$.
- $$= \frac{4x^2y^2}{x^2y^4} = 4x^{2-2}y^{2-4} = 4y^{-2}$$

- 4 Simplify the following, giving your answer in index form:

$$7^7 \times 7^9 = 7^{7+9} = 7^{16}$$

- 5 Simplify the following using Index Laws:

$$\begin{aligned} &(-6y^4) \times (-6y^3) \times (-4y^2) \\ &[(-6) \times (-6) \times (-4)] \times y^{4+3+2} \\ &-144y^9 \end{aligned}$$

- 6 Simplify $5^{2m} \times 5^{m+1}$

$$5^{2m+m+1} = 5^{3m+1}$$

- 7 Simplify the following, giving your answer in index form:

$$\frac{27x^9y^9}{3x^2y^6} = \frac{27x^{9-2}y^{9-6}}{3} = \frac{27x^7y^3}{3} = 9x^7y^3$$

- 8 Simplify the following, giving your answer in index form:

$$\frac{u^{3x+1}}{u^x} = u^{3x+1-x} = u^{2x+1}$$

Collect like terms

Chapter 2 Algebra 1

$$7a + 5 - 4a = \underline{3a + 5}$$

$$3d \times 5e = 3 \times d \times 5 \times e = \underline{15de}$$

$$5mn \times^{-2}m^2n = 5 \times^{-2}m \times m^2 \times n \times n = \underline{-10m^3n^2}$$

$$\frac{18ay \div 4a}{4a} = \frac{18ay}{4a} = \underline{\frac{9y}{2}}$$

Distributive Law: $a(b + c) = ab + ac$

$$3(2b - 5) = \underline{6b - 15}$$

$$-4(x - 1) - 2(x - 4) = -4x + 4 - 2x + 8 = \underline{-6x + 12}$$

Factorisation: $a(b + c) = ab + ac$

$$4x - 12 = \underline{4(x - 3)}$$

$$-6x^2 + 15x = \underline{-3x(2x - 5)}$$

Distributive Law

- 1 Remove the brackets and simplify.

$$8(u + v) = \underline{8u + 8v}$$

- 2 Simplify:
 $7(y + 3) + 2y + 6$

$$7y + 21 + 2y + 6 = \underline{9y + 27}$$

- 3 Simplify:
 $6y + 3(y + 7) - 7$

$$6y + 3y + 21 - 7 = \underline{9y + 14}$$

4

Expand and simplify:

$$5(y - 7) + 6(y + 5)$$

$$5y - 35 + 6y + 30$$

$$11y - 5$$

5

Expand and simplify:

$$m(m + 7) - 6(m + 5)$$

$$m^2 + 7m - 6m - 30$$

$$m^2 + 1m - 30$$

6

Expand and simplify $2a(a + 9b - 4c)$.

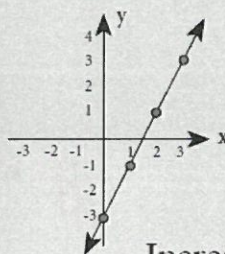
$$2a^2 + 18ab - 8ac$$

Chapter 4 Linear & Non-Linear Graphs

Linear Graphs are of the form:

$$y = mx + c$$

- a) m is the increase each step.
b) c is the value of y when $x = 0$ ie $(0, c)$.

Increase of 2 each $x \rightarrow 2x$ When $x=0$, $y=-3 \rightarrow y = 2x - 3$

Linear Relationships

1

1	x	0	1	2	3	4
	y	-2	1	4	7	10

$$+3 \quad +3 \quad +3 \quad +3$$

At $x = 0$

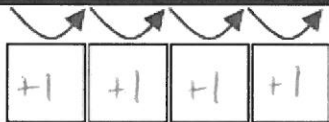
$$y = -2$$

$$\text{Gradient: } +3$$

2

4

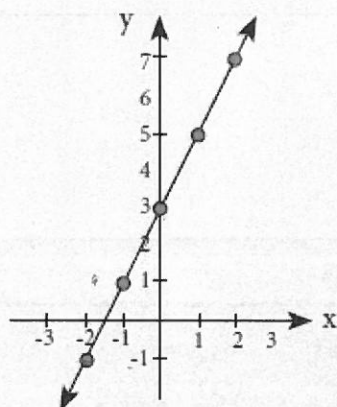
x	-2	-1	0	1	2
y	5	6	7	8	9

At $x = 0$ $y = 7$ Gradient: $+1$

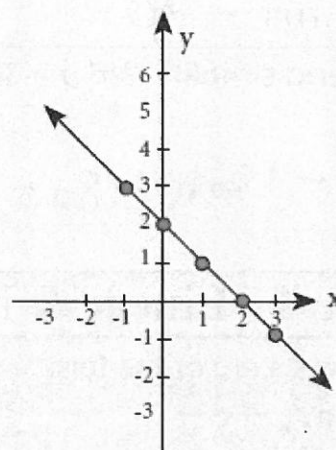
3

Write a rule for each of the following graphs:

a)



b)

 $\frac{\text{rise}}{\text{run}}$

$$\frac{2}{1} = 2$$

$$c = 3$$

$$y = 2x + 3$$

$$y = mx + c$$

$$\frac{-1}{1}$$

$$m = -1$$

$$c = 2$$

$$y = -1x + 2$$